

C880

Algebra for Secondary Mathematics Teaching

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A teacher who mentors two new colleagues gives the problem $8 + 4 = __ + 5$ to a class of 27. Many students answer 12. What does this answer most likely reveal?

- A. Students understand equality but rushed.
- B. Students read the equal sign as “write the answer,” not as “same value as.”
- C. Students added all the numbers.
- D. Students made a subtraction error.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

At a university with 6 times as many students as professors, a teacher who has taught Algebra I for eight years asks students to write an equation using S for students and P for professors. Many write $6S = P$. The university has a large library. What is the most effective response?

- A. Tell students to write the equation in the same order as the words.
- B. Replace S and P with x and y so the letters carry no meaning.
- C. Have students test the equation with numbers, such as 10 professors, to see that $6S = P$ gives fewer students than professors.
- D. Accept $6S = P$, since it uses the right numbers.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A recipe uses 4 cups of flour for 6 cups of milk. A student in a class that meets right after lunch says 8 cups of flour need 10 cups of milk, “because you add 4 to both.” The recipe is from a cookbook printed in 1980. What misconception does this show?

- A. Additive reasoning in a multiplicative situation; the ratio requires 12 cups of milk.
- B. Old recipes use different measuring cups.
- C. A computation error; $8 + 4 = 12$, not 10.
- D. The student is right, since both amounts increased.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Which relationship is proportional?

- A. $y = 3x$
- B. $y = x^2$
- C. Any relationship whose graph is a straight line
- D. $y = 3x + 2$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A teacher who uses graphing software every day shows a table: $x = 0, 1, 2, 3$ and $y = 5, 10, 20, 40$. The software license was renewed in June. How should the relationship be classified?

- A. Exponential with growth factor 5
- B. Exponential, because y doubles for each increase of 1 in x
- C. Linear with slope 5
- D. Linear, because y increases every time

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A function has $y = 7$ at $x = 0$ and increases by 3 each time x increases by 1. Which equation models it?

- A. $y = 7 \cdot 3^x$
- B. $y = 3 + 7x$
- C. $y = 3x$
- D. $y = 7 + 3x$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A teacher reviewing work from a summer bridge program writes $f(x) = 2x + 3$ on the board. A student writes $f^{-1}(x) = 1/(2x + 3)$. The program met four days a week. What is the correct inverse and the best explanation?

- A. The student is right; -1 means reciprocal.
- B. $f^{-1}(x) = -2x - 3$
- C. $f^{-1}(x) = (x - 3)/2$, because the inverse undoes adding 3 and then undoes multiplying by 2
- D. $f^{-1}(x) = (x/2) - 3$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

If $f(1) = 5$ and $f(5) = 17$, what is $f^{-1}(5)$?

- A.** $1/5$
- B.** 1
- C.** 5
- D.** 17

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Grading a quiz during a planning period that lasts 50 minutes, a teacher sees a student simplify $(x + 6)/6$ to x . The quiz had 15 items. What is the best feedback?

- A.** Tell the student never to cancel anything in a fraction.
- B.** Mark it wrong and write the correct answer, $x/6 + 1$.
- C.** Accept it, because it works when $x = 0$.
- D.** Ask the student to substitute $x = 6$ into both expressions (2 vs. 6), then discuss why only factors, not terms, can be divided out.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A teacher, in a room decorated with student posters, poses: "A ribbon is $3/4$ m long. How many $1/8$ m pieces can be cut from it?" A student says the answer must be less than $3/4$ "because division makes things smaller." The ribbon is red. What is the answer, and what idea should the teacher build?

- A.** $3/32$ m; the student is right that division makes smaller.
- B.** $1/6$ of a piece
- C.** 6 pieces, because you always flip and multiply.
- D.** 6 pieces; dividing by a number less than 1 gives a quotient larger than the dividend.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A teacher, whose class is using new graphing calculators, shows $f(x) = (x - 2)(x + 3)$. The calculators came in a set of 30. A student says the x-intercepts are at 2 and 3. What is correct?

- A.** The x-intercepts are (2, 0) and (3, 0).
- B.** The x-intercepts are (−2, 0) and (3, 0).
- C.** The x-intercepts are (2, 0) and (−3, 0).
- D.** The y-intercept is −6, so there are no x-intercepts.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

12. [TRANSFER — reversed stem / different variable]

A quadratic has x-intercepts at −1 and 4 and leading coefficient 1. Which function is it?

- A.** $f(x) = (x + 1)(x - 4)$
- B.** $f(x) = (x - 1)(x + 4)$
- C.** $f(x) = x^2 - 4$
- D.** $f(x) = (x + 1) + (x - 4)$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A teacher ~~who mentors two new colleagues~~ gives the problem $8 + 4 = _ + 5$ ~~to a class of 27~~. Many students answer 12. What does this answer most likely reveal?

Target word: diagnose the error **Decides:** 12 is the sum of the left side, written as “the answer.”

A	Absolute language	A widespread consistent answer signals a misconception, not carelessness.
B	Correct	They computed $8 + 4$ and ignored the $+ 5$; the correct answer is 7.
C	Confusion set	That error produces 17, not 12.
D	Adjacent category	No subtraction was attempted; the error is conceptual.

2. Answer C Concept 2

At a university with 6 times as many students as professors, a teacher ~~who has taught Algebra I for eight years~~ asks students to write an equation using S for students and P for professors. Many write $6S = P$. ~~The university has a large library.~~ What is the most effective response?

Target word: respond to the reversal error **Decides:** Students treat S as a label for “students,” not a number.

A	Inverted operation	Word-order matching is what causes the reversal.
B	Adjacent category	Hides the issue without building the idea of a variable as a quantity.
C	Correct	Testing values shows the letters stand for quantities and exposes the reversal.
D	Dropped element	Ignores that the relationship is backward.

3. Answer A Concept 3

A recipe uses 4 cups of flour for 6 cups of milk. A student ~~in a class that meets right after lunch~~ says 8 cups of flour need 10 cups of milk, “because you add 4 to both.” ~~The recipe is from a cookbook printed in 1980.~~ What misconception does this show?

Target word: name the misconception **Decides:** The student added instead of scaling.

A	Correct	4:6 scaled by 2 is 8:12.
B	Camouflage	The print date does not decide the ratio.
C	Confusion set	The student’s arithmetic matches their method; the method is wrong.
D	Absolute language	Increasing both terms does not keep the ratio.

4. Answer A Concept 3 • transfer

Which relationship is proportional?

Target word: identify proportionality **Decides:** Constant ratio y/x and passes through the origin.

A	Correct	$y/x = 3$ for every $x \neq 0$.
B	Confusion set	The ratio y/x changes with x .
C	Absolute language	Only lines through the origin are proportional.
D	Adjacent category	Linear but not proportional: the intercept is 2.

5. Answer B Concept 4

A teacher ~~who uses graphing software every day~~ shows a table: $x = 0, 1, 2, 3$ and $y = 5, 10, 20, 40$. ~~The software license was renewed in June.~~ How should the relationship be classified?

Target word: linear or exponential **Decides:** Each step multiplies y by 2.

A	Wrong stem number	5 is the starting value; the factor is 2.
B	Correct	Constant ratio per equal step in x .
C	Dropped element	Uses only the first difference.
D	Confusion set	Increasing is not the test; differences (5, 10, 20) are not constant.

6. Answer D Concept 4 • transfer

A function has $y = 7$ at $x = 0$ and increases by 3 each time x increases by 1. Which equation models it?

Target word: build the function from covariation **Decides:** Constant difference = linear.

A	Confusion set	Multiplies by 3 per step: exponential.
B	Inverted operation	Swaps the starting value and the rate.
C	Dropped element	Leaves out the starting value 7.
D	Correct	Start 7, add 3 per unit step.

7. Answer C Concept 5

A teacher reviewing work ~~from a summer bridge program~~ writes $f(x) = 2x + 3$ on the board. A student writes $f^{-1}(x) = 1/(2x + 3)$. ~~The program met four days a week.~~ What is the correct inverse and the best explanation?

Target word: correct inverse **Decides:** Undo “multiply by 2, then add 3.”

A	Confusion set	For functions, -1 denotes the inverse, not the reciprocal.
B	Adjacent category	Negates the function instead of inverting it.
C	Correct	Reverse operations in reverse order.
D	Inverted operation	Undoes the steps in the wrong order.

8. Answer B Concept 5 • transfer

If $f(1) = 5$ and $f(5) = 17$, what is $f^{-1}(5)$?

Target word: evaluate an inverse **Decides:** f^{-1} swaps input and output.

A	Inverted operation	Treats the inverse as a reciprocal.
B	Correct	$f(1) = 5$, so $f^{-1}(5) = 1$.
C	Wrong stem number	Returns the input unchanged.
D	Confusion set	That is $f(5)$, not $f^{-1}(5)$.

9. Answer D Concept 6

Grading a quiz during a planning period that lasts 50 minutes, a teacher sees a student simplify $(x + 6)/6$ to x . The quiz had 15 items. What is the best feedback?

Target word: feedback on a false simplification **Decides:** Only common factors cancel; 6 is a term.

A	Adjacent category	Overcorrects; canceling common factors is valid.
B	Dropped element	Gives the answer without addressing why the cancellation fails.
C	Absolute language	Equivalence must hold for all values; one match proves nothing.
D	Correct	A test value exposes non-equivalence and leads to the rule's reason.

10. Answer D Concept 7

A teacher, in a room decorated with student posters, poses: "A ribbon is $3/4$ m long. How many $1/8$ m pieces can be cut from it?" A student says the answer must be less than $3/4$ "because division makes things smaller." The ribbon is red. What is the answer, and what idea should the teacher build?

Target word: answer and concept **Decides:** $3/4 \div 1/8$ asks how many eighths are in three-fourths.

A	Inverted operation	Multiplies instead of dividing.
B	Confusion set	Divides $1/8$ by $3/4$, reversing dividend and divisor.
C	Dropped element	Right number, but gives the procedure without the meaning the teacher needs to build.
D	Correct	$3/4 = 6/8$, so six eighths fit.

11. Answer C Concept 8

A teacher, whose class is using new graphing calculators, shows $f(x) = (x - 2)(x + 3)$. The calculators came in a set of 30. A student says the x-intercepts are at 2 and 3. What is correct?

Target word: zeros from factors **Decides:** $x + 3 = 0$ gives $x = -3$.

A	Inverted operation	Sign error on the second factor.
B	Inverted operation	Flips both signs.
C	Correct	Each factor equals zero at its root.
D	Adjacent category	The y-intercept is -6 , but that does not remove the x-intercepts.

12. Answer A Concept 8 · transfer

A quadratic has x-intercepts at -1 and 4 and leading coefficient 1. Which function is it?

Target word: build from intercepts **Decides:** Zero r gives factor $(x - r)$.

A	Correct	Zeros at -1 and 4 .
B	Inverted operation	Sign error: zeros at 1 and -4 .
C	Confusion set	Zeros at ± 2 .
D	Dropped element	Adds the factors instead of multiplying; that is linear.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	The equal sign as a relation	The equal sign means both sides have the same value, not “write the answer next.”	Reading = as an instruction to compute, which leads to answers like 12 or 17 for $8 + 4 = _ + 5$.	An open number sentence with operations on both sides.
2	Variables represent quantities	A letter stands for a number (a quantity), not a label or object; equations relate quantities.	The reversal error: writing $6S = P$ for “six times as many students as professors.”	A word statement comparing two quantities to be written as an equation.
3	Proportional vs. additive reasoning	A proportional relationship has a constant ratio ($y = kx$, through the origin); equivalent ratios scale multiplicatively.	Adding the same amount to both terms of a ratio, or calling any linear function proportional.	A missing-value ratio problem, or a line with a nonzero intercept.
4	Covariation: linear vs. exponential change	Linear functions add a constant difference per equal step in x ; exponential functions multiply by a constant ratio per equal step.	Classifying a table by whether it “goes up fast” instead of by differences vs. ratios.	A table with equally spaced x -values.
5	Inverse functions	f^{-1} undoes f : if $f(a) = b$ then $f^{-1}(b) = a$; it swaps inputs and outputs and is not $1/f$.	Reading the -1 as an exponent (reciprocal).	Notation f^{-1} , or “undo” a function.
6	Equivalent expressions	Two expressions are equivalent only if they agree for every value of the variable; testing a value can quickly show non-equivalence.	“Canceling” terms across a sum, such as $(x + 6)/6 = x$, or $(a + b)^2 = a^2 + b^2$.	A student simplification that removes terms rather than factors.
7	Meaning of fraction division	$a \div b$ asks how many b ’s fit in a (or how much per one b), so dividing by a number less than 1 gives a larger result.	Believing division always makes smaller, or “flip” without meaning.	Dividing by a fraction less than 1 in a context.
8	Zeros, factors, and x-intercepts	If $(x - r)$ is a factor, then r is a zero and $(r, 0)$ is an x -intercept; the three representations are linked.	Sign errors: reading $(x + 3)$ as giving a zero at $+3$.	A factored polynomial and a graph or zeros question.

Course scope (public description): Algebra for Secondary Mathematics Teaching explores important conceptual underpinnings, common misconceptions and students’ ways of thinking, appropriate use of technology, and instructional practices to support and assess the learning of algebra. Secondary teachers should have an understanding of the following: algebra as an extension of number, operation, and quantity; various ideas of equivalence as it pertains to algebraic structures; patterns of change as covariation between quantities; connections between representations (tables, graphs, equations, geometric models, context); and the historical development of content and perspectives from diverse cultures. In particular, the course focuses on deeper understanding of rational numbers, ratios and proportions, meaning and use of variables, functions (e.g., exponential, logarithmic, polynomials, rational, quadratic), and inverses. Calculus I is a prerequisite for this course.